

Odisha Mining Corporation (OMC)

Advanced AI Chatbot Knowledge Base Document

This knowledge base document is designed for the Odisha Mining Corporation AI-powered conversational chatbot system. The chatbot uses Retrieval-Augmented Generation (RAG), semantic document search, conversational AI, and enterprise support workflows to assist vendors, employees, transporters, contractors, and citizens interacting with OMC services.

Vendor Registration & Compliance

Vendors participating in Odisha Mining Corporation procurement activities must complete the registration process through the official vendor onboarding system. Vendors are required to submit GST registration certificates, PAN card details, financial statements, compliance declarations, environmental clearances where applicable, and supporting operational documentation. The AI chatbot should assist vendors in understanding document submission workflows, compliance policies, approval timelines, and vendor category selection processes.

Mining Tender Participation Guidelines

Odisha Mining Corporation publishes multiple tenders related to transportation, mining operations, equipment procurement, infrastructure development, safety management, and digital transformation projects. Vendors participating in tenders must review eligibility requirements, EMD submission guidelines, technical qualification criteria, and financial bid documentation carefully before submission. The AI chatbot should intelligently answer questions related to tender participation workflows, document requirements, bid submission timelines, and procurement compliance procedures.

Transport & Dispatch Operations

Transport vendors involved in mineral dispatch operations must maintain valid route permissions, vehicle authorization documents, dispatch permits, mineral transit passes, and weighbridge approvals. Transport operations are monitored for compliance with mining safety and environmental regulations. The AI chatbot should help transport operators check dispatch status, route permissions, vehicle verification details, and document compliance requirements using integrated workflows.

Mining Safety & Environmental Compliance

Mining operations require strict adherence to environmental and occupational safety standards. All contractors, operators, and vendors must comply with PPE usage policies, safety audit procedures, emergency response protocols, hazardous material handling standards, and environmental monitoring regulations. The AI chatbot should provide contextual responses related to mining safety procedures, environmental compliance guidelines, inspection protocols, and incident escalation workflows.

Employee HR & Welfare Policies

Odisha Mining Corporation employees are entitled to multiple HR and welfare services including leave management, medical benefits, travel allowances, payroll support, pension guidance, grievance handling, safety training programs, and internal policy assistance. The chatbot should provide employees with quick access to HR policies, training schedules, leave rules, TA/DA guidelines, and grievance escalation mechanisms through conversational workflows.

AI Chatbot Operational Knowledge Base

AI-Powered Semantic Search

The OMC AI chatbot uses semantic search and vector embeddings to retrieve accurate contextual information from uploaded PDF documents, policy manuals, operational SOPs, tender files, HR guidelines, compliance documents, and enterprise knowledge repositories. Instead of traditional keyword search, the chatbot understands the meaning and intent behind user queries and retrieves the most relevant information using vector similarity search techniques.

Human Escalation & Support Workflow

When the chatbot cannot confidently resolve user issues, the system should intelligently escalate requests to appropriate support officers or departments. The escalation workflow includes ticket generation, issue categorization, conversation history transfer, support officer assignment, and ticket status tracking. The chatbot should maintain professional government-style communication throughout the escalation process.

Document Intelligence & RAG Pipeline

Uploaded PDF documents are processed using a Retrieval-Augmented Generation pipeline consisting of document ingestion, OCR-ready parsing, chunk generation, vector embedding creation, metadata indexing, and semantic retrieval. The chatbot should retrieve contextual chunks from indexed documents and generate grounded AI responses while minimizing hallucinations and misinformation.

Enterprise Integration Architecture

The AI chatbot platform should support enterprise integrations with ERP systems, CRM applications, vendor management systems, SAP workflows, transport management platforms, complaint management systems, and internal operational dashboards. These integrations enable real-time retrieval of vendor status, payment information, dispatch records, employee services, and tender updates.

Future AI Enhancement Roadmap

Future enhancements for the OMC AI chatbot include multilingual support, voice-based conversational AI, WhatsApp integration, OCR invoice extraction, AI analytics dashboards, predictive vendor recommendations, automated compliance monitoring, document summarization, smart escalation routing, and mining operation insights. These capabilities will support Odisha Mining Corporation's long-term digital transformation strategy.